

Olist Customer Experience

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Source snapshot analysis. Values use original historical dataset periods.

Problem

Which delivery stages, routes and seller groups are associated with poor reviews and unreliable delivery?

Method

Preserve raw snapshots, validate source keys and foreign keys, create grain-safe analytical tables, execute SQLite analyses, reconcile totals, and review descriptive Python statistics. Excel supports comparison and prioritization; the Power BI package specifies reproducible visuals and measures.

Most important findings

99,441 orders; 96,478 delivered; 625 cancelled (0.63%).

96,470 orders had valid delivery and estimated dates; 6.77% were late by calendar date.

Mean review was 4.28 for on-time orders versus 2.26 for late orders; association does not establish causation.

Mean processing time was 2.85 days and mean shipping time was 9.33 days; available-timestamp samples differ.

170 sellers met the priority rule: at least 50 eligible orders and a late rate above the portfolio rate.

Business impact

The analysis identifies where to investigate and test operational or commercial changes. Financial savings and causal impact are not estimated without intervention costs and experimental evidence.

Recommendations

Audit high-volume sellers with above-baseline late rates first; verify carrier handoff records and inventory availability before attributing responsibility.

Compare processing and shipping separately on the same complete-timestamp sample before assigning an operational intervention.

Test revised delivery estimates on routes with persistent positive delays; track both promise accuracy and total transit time.

Use matched-route comparisons or controlled operational experiments to evaluate improvements in reviews.

Measurement limitations

Calendar dates define on-time delivery because estimated dates are day-level promises. Same-day delivery counts as on-time.

Latest review response, then creation time and review ID, determines one selected review per order. Raw review history remains unchanged.

Processing and shipping averages exclude negative timestamp intervals independently; their samples may differ.

Multi-seller and multi-category orders count once at order grain. Seller/category comparisons use distinct order membership and cannot uniquely assign blame.

